

Prof. Siddhartha Srinivasa

The Personal Robotics Lab
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Employment

Professor	2023-
Boeing Endowed Professor in Computer Science & Engineering	2017-23
Computer Science & Engineering Department, University of Washington	
Finmeccanica Associate Professor in Computer Science	2013-17
Associate Professor	2011-13
The Robotics Institute, Carnegie Mellon University	
Venture Partner, Madrona Venture Group	2025-
Member, Board of Directors, Zordi Inc.	2021-26
Distinguished Engineer, Cruise Inc.	2022-25
Director, Robotics AI, Amazon Inc.	2018-22
First Wave Founder, Berkshire Grey Inc.	2014-18
Senior Research Scientist, Intel Labs Pittsburgh	2005-11

Education

Ph.D., Carnegie Mellon University (CMU)	August 2005
Advisors: Michael Erdmann & Matthew Mason	Thesis: <i>Control Synthesis for Dynamic Contact Manipulation</i>
B. Tech., Indian Institute of Technology Madras (IITM)	August 1999
Advisor: A. Radhakrishnan	Thesis: <i>Reverse Engineering using the Structured Lighting Technique</i>

Honors and Awards

IEEE Fellow	2018
Boeing Endowed Professorship in Computer Science	2017-23
CMU Women's Association outstanding graduating senior advisor (Rachel Holladay)	2017
Finmeccanica Chair in Computer Science	2013-16
RSS Early Career Spotlight Award	2013
Robotics Institute Cool Person of the Year Award	2012
Okawa Foundation Research Grant	2012
Office of Naval Research Young Investigator Award	2012
RAS Most Active Technical Committee Award: Mobile Manipulation	2011
Intel Corporate Technology Group Divisional Recognition Awards	2008
Intel Research Pittsburgh Lab Research Awards	2006-07
Jawaharlal Nehru Summer Research Fellowship, Indian Institute of Science	1997-98
Rajiv Gandhi Award for Best Summer Research Fellow, Indian Institute of Science	1997-98
Indian National Mathematics Olympiad	1994

Mentoring

Current Ph.D. Students

Milad Mesbahi	2026-
Ethan Pronovost	2024-

Yunchu Zhang (+Abhishek Gupta)	2023-
Sidharth Talia	2022-
Helen Wang (+Dieter Fox)	2022-

Alumni - Postdoctoral Fellows

Taylor Kessler Faulkner	2022-2024
	Lecturer @ University of Washington
Christoforos Mavrogiannis	2019-2023
	Assistant Professor @ Michigan
Sanjiban Choudhury	2018-2019
	Assistant Professor @ Cornell
Matthew Barnes	2018-2019
	Researcher @ Google
Tapomayukh Bhattacharjee	2017-2021
	Assistant Professor @ Cornell
Oren Salzman	2016-2019
	Assistant Professor @ Technion
Daqing Yi	2016-2018
	Researcher @ Google
Henny Admoni	2015-2017
	Assistant Professor @ CMU
Aaron Johnson	2015-2016
	Assistant Professor @ CMU
Prasanna Velagapudi	2012-2014
	CTO @ Agility

Alumni - Ph.D.

Amal Nanavati (+Maya Cakmak)	2019-2025
<i>Towards In-Home Deployments of Physically Assistive Robots: Insights from Robot-Assisted Feeding for People with Motor Impairments</i>	Researcher @ Zoox
Matthew Schmitt	2018-2025
<i>Off-Road Navigation Under Sensing Uncertainty</i>	Researcher @ Overland AI
Ethan Gordon	2018-2024
<i>Tractably Adaptable Food Manipulation for Robot-Assisted Feeding</i>	Postdoc @ Penn
Liyiming Ke (+Abhishek Gupta)	2017-2024
<i>Teach Robot to Use Chopsticks: a Test Bed for Robotic Learning via Fine Manipulation</i>	Researcher @ Physical Intelligence
William Agnew	2017-2023
<i>Human Priors for Reinforcement Learning</i>	Postdoc @ CMU
Sherdil Niyaz	2017-2022
<i>Optimizing the Design of Robot Environments via Interleaved Optimization and White-Box Motion-Planning</i>	Researcher @ Waymo
Brian Hou	2016-2022
<i>Robot Motion Planning with Uncertainty and Urgency</i>	Researcher @ Aurora
Samuel Ainsworth	2016-2022
<i>Perspectives on Policy Learning</i>	Researcher @ Waymo
Patrick Lancaster (+Josh Smith)	2014-2022
<i>Towards Dexterous In-Hand Manipulation via Electrostatic Braking and Pre-touch Sensing</i>	Postdoc @ Meta

Aditya Vamsikrishna <i>Efficient Robot Motion Planning in Cluttered Environments</i>	2016-2021 Researcher @ Aurora
Gilwoo Lee <i>Scalable Bayesian Reinforcement Learning</i>	2015-2020 Founder @ Zordi
Stefanos Nikolaidis <i>Mathematical Models of Adaptation in Human-Robot Collaboration</i>	2014-2018 Assistant Professor @ USC
Laura Herlant <i>Algorithms and Implementation and Studies on Eating with a Shared Control Robot Arm</i> Robotics and AI Institute	2013-2018 Director of Robot Perception @
Zita Marinho (+Geoff Gordon) <i>Moment-based Algorithms for Structured Prediction</i>	2012-2018 Research Scientist @ DeepMind
Shervin Javdani (+Drew Bagnell) <i>Acting under Uncertainty for Information Gathering and Shared Autonomy</i>	2011-2017 Researcher @ Aurora
Matt Klingensmith (+Michael Kaess) <i>Tracking and Calibrating Robot Arms using SLAM Techniques</i>	2013-2016 Autonomy Engineer @ Wayve
Jennifer King <i>Robust Rearrangement Planning using Nonprehensile Interaction</i>	2012-2016 Researcher @ Berkshire Grey
Michael Koval (+Nancy Pollard) <i>Robust Manipulation via Contact Sensing</i>	2012-2016 Researcher @ Waymo
Christopher Dellin <i>Completing Manipulation Tasks Efficiently in Complex Environments</i>	2009-2016 Robotician @ Nuro
Anca Dragan <i>Legible Robot Motion Planning</i>	2009-2015 Associate Professor @ Berkeley
Mehmet Dogar <i>A Framework for Manipulation in Cluttered Environments</i>	2008-2013 Associate Professor @ Leeds
Alvaro Collet (+Martial Hebert) <i>Lifelong Robotic Object Perception</i>	2009-2012 Engineering Lead @ Facebook
Dmitry Berenson (+James Kuffner) <i>Constrained Manipulation Planning</i>	2006-2011 Associate Professor @ Michigan

Alumni - M.S.

Jose Jaime <i>Extending Robot-Assisted Feeding to Spoon-Based Foods with the Articutool: A Modular Active End-Effector</i> Velodex AI	2023-2025 Engineer @
Atharva Pradhan <i>Insights from Developing a Robotic Bimanual Manipulation System</i>	2023-2025 Engineer @ Third Wave Automation
Jeongseok Lee <i>A Linear-Time Variational Integrator for Multibody Systems</i>	2016-2018 Researcher @ Meta

Pengju Jin	2017-2017
<i>Highly Robust Pose Estimation from Single Frame RGBD</i>	Researcher @ Aurora
Shushman Choudhury	2015-2017
<i>Anytime Geometric Motion Planning on Large Dense Roadmaps</i>	AI Software Engineer @ Google Research
Rosario Scalise (+Stephanie Rosenthal)	2015-2017
<i>Human-Centered Design of Robot Explanations</i>	Ph.D. @ UW
Shen Li (+Stephanie Rosenthal)	2015-2017
<i>Automatically Evaluating and Generating Clear Robot Explanations</i>	Ph.D. @ MIT
Evan Shapiro	2013-2015
<i>A Hierarchical Framework for Configuration Space Task Planning</i>	CEO @ Mina Foundation
Aaron Walsman	2013-2015
<i>ROCK: Robust Object Constellation for Kinematic Pose</i>	Postdoc @ Harvard
Elizabeth Cha (+Jodi Forlizzi)	2012-2014
<i>Robots in the Home: Qualitative and Quantitative Insights into Kitchen Organization</i>	Researcher @ Waymo
Kyle Strabala	2010-2012
<i>Learning the Communication of Intent Prior to Physical Collaboration</i>	Scientist @ Near Earth Autonomy
Martin Herrmann (+Dr.-Inf. Uwe Hanebeck @ Universitat Karlsruhe)	2009-2009
<i>Active scene and object reconstruction for robotic manipulation from vision and laser</i>	TU Braunschweig
Garratt Gallagher (+Drew Bagnell)	2007-2009
<i>GATMO: A Generalized Approach to Tracking Movable Objects</i>	Google Robotics

Alumni - Other

Ajinkya Kamat, Staff	2018-2019
Research: <i>Outdoor Unstructured Mobile Manipulation</i>	MRSD @ CMU
Youngsun Kim, Staff	2017-2019
Research: <i>Robot-Assisted Feeding</i>	Engineer @ Zordi
Hanjun Song, Staff	2016-2019
Research: <i>Sensing Shear Forces During Food Manipulation</i>	Ph.D. @ MIT
Rachel Holladay, B.S.	2013-2017
Thesis: <i>Following Paths in Task Space: Distance Metrics and Planning Algorithms</i>	Assistant Professor @ Penn
Pyry Matikainen, Teaching Fellow	2015-2017
Research: <i>Visual Computing</i>	

Graduate Interns

Atharva Pradhan (UW)	2024
Jonathan Spencer (Princeton)	2022
Yunchu Zhang (CMU)	2022
Rishabh Madan (IIT Kharagpur)	2020
Sara Sheikholeslami (UBC)	2019
Lerrel Pinto (CMU)	2019

Daniel Gallenberger (TU Munich)	2018
Stefania Pellegrinelli (ITIA-CNR)	2015
Marco Cagnetti (University of Rome)	2015
Joshua Haustein (Universitat Karlsruhe)	2014
Henny Admoni (Yale)	2013
Georg Bartels (TU Bremen)	2013
Katie Correll (CMU)	2013
Michael Koval (CMU)	2012
Steven Gray (Penn)	2011
Vincenzo Micelli (University of Parma)	2011
Tim Niemueller (RWTH Aachen)	2010
Peter Kaiser (Universitat Karlsruhe)	2010
Maya Cakmak (Georgia Tech)	2010
Alex Sorokin (UIUC)	2010
Lillian Chang (CMU)	2009
Ross Knepper (CMU)	2009
Laura Lindzey (CMU)	2009
Manel Martinez (CMU)	2009
Julius Ziegler (Universitat Karlsruhe)	2009
Nico Blodow (TU Munich)	2009
Rosen Diankov (CMU)	2009
Geoffrey Hollinger (CMU)	2008
Kevin Peterson (CMU)	2008
Nathan Ratliff (CMU)	2008
Martin Rufli (ETH Zurich)	2008
Michael Ashley-Rollman (CMU)	2006
Stuart Anderson (CMU)	2006
Michael DeRosa (CMU)	2006
Ashish Deshpande (Michigan)	2006
Jiaxin Fu (CMU)	2006
Preethi Bhatt (CMU)	2006

Undergraduate Interns

Abhay Deshpande (UW)	2022-2024
Quanquan Peng (SJTU)	2024
Arnav Thareja (UW)	2023
Rohan Baijal (IIT Kanpur)	2022-2023
Sriyash Poddar (IIT Kharagpur)	2022-2023
Ramya Challa (UW)	2022-2023
Sumegh Roychowdhury (IIT Kharagpur)	2021
Colin Summers (UW)	2018-2020
Matthew Rockett (UW)	2018-2020
Sumegh Roychowdhury (IIT Kharagpur)	2020
Sidharth Talia (Bharati Vidyapeeth College of Engineering)	2020
Johan Michalov (UW)	2018-2019
Rajat Kumar Jenamani (IIT Kharagpur)	2019
Jeffrey Maxwell (UW)	2019
Shivam Singhal (UW)	2019
Minae Kwon (Stanford)	2018
Savanna Yee (UW)	2018
Nanda Sundaresan (UW)	2018
Kaiden James Field (UW)	2018
Connor Geiman (UW)	2018
Tao Jin (UW)	2018
Rahul Vernwal (IIT Kharagpur)	2018
Maha Alrashed (BU)	2018

Abdullah Albakry (NC State)	2018
Ramon Qu (UW)	2017
Nanda Sundaresan (UW)	2017
Jeffrey Maxwell (UW)	2017
Vinitha Ranganeni (CMU)	2015-2017
Kevin Zhang (CMU)	2015-2017
Pengju Jin (CMU)	2014-2017
Rachel Holladay (CMU)	2013-2017
Joey Fernau (CMU)	2015
Angela Wang (CMU)	2014
Peter McHale (CMU)	2014
Vikram Sunder (CMU)	2014
Don Zheng (CMU)	2014
Neil Jassal (CMU)	2014
Myles Blodnick (CMU)	2014
Scott Martin (CMU)	2014
Yuyang Guo (CMU)	2013-2014
Andrey Kurenkov (Georgia Tech)	2013
Abhijeet Tallavajhula (IIT)	2012
Bo Xiong (Connecticut)	2012
Corina Guarau (Jacobs)	2012
Debidatta Dwebi (IIT)	2012
Kenton Lee (Penn)	2012
Dominic Zirbel (CMU)	2012
Alex Zirbel (CMU)	2011-2012
Tom Mullins (CMU)	2011
Nick Stanley (CMU)	2011
Tudor Achim (CMU)	2009-2010
Andrew Yeager (CMU)	2009
Ian-Clanton Thuon (CMU)	2008-2009
Daniel Dewey (CMU)	2007-2008

Ph.D Thesis Committees

Vinitha Ranganeni (UW)	2024
Nick Walker (UW)	2024
Nathan Hatch (UW)	2024
Mohak Bharadwaj (UW)	2024
Anqi Li (UW)	2024
Ekta Samani (UW)	2023
Christopher Xie (UW)	2021
Senka Krivic (University of Innsbruck)	2019
Parker Owan (UW)	2019
Arunkumar Byravan (UW)	2019
Rahul Warrier (UW)	2018
Justin Huang (UW)	2018
Connor Schenk (UW)	2017-18
Kiril Solovey (Technion)	2018
Sanjiban Choudhury (CMU)	2013-17
Venkatraman Narayanan (CMU)	2013-17
Breelyn Kane Styler (CMU)	2011-18
Mike Phillips (CMU)	2011-15
Alberto Rodriguez (CMU)	2007-13
Ross Knepper (CMU)	2006-11
Nathan Ratliff (CMU)	2004-09

Refereed Journals

- [1] Vincent Roulet, S.S. Srinivasa, Maryam Fazel, and Zaid Harchaoui. On global and local convergence of iterative linear quadratic optimization algorithms for discrete time nonlinear control. *Journal of Machine Learning Research*, 2025.
- [2] S. Talia, O. Salzman, and S. S. Srinivasa. Incremental Generalized Hybrid A*. *IEEE Robotics and Automation Letters*, 2025.
- [3] Patrick Lancaster, Christoforos Mavrogiannis, Siddhartha Srinivasa, and Joshua R. Smith. Electrostatic brakes enable individual joint control of underactuated, highly articulated robots. *The International Journal of Robotics Research*, 2024.
- [4] Jaein Lim, Mahdi Ghanei, Connor Lawson, Siddhartha Srinivasa, and Panagiotis Tsiotras. Lazy incremental search for efficient replanning with bounded suboptimality guarantees. *The International Journal of Robotics Research*, 2024.
- [5] Vincent Roulet, S.S. Srinivasa, Maryam Fazel, and Zaid Harchaoui. Iterative linear quadratic optimization for nonlinear control: Differentiable programming algorithmic templates. *Open Journal of Mathematical Optimization*, 5:1–63, 2024.
- [6] C Mavrogiannis, K Balasubramanian, S. Poddar, A. Gandra, and S. S. Srinivasa. Winding through: Crowd navigation via topological invariance. *IEEE Robotics and Automation Letters*, 8(1), 2023.
- [7] Christoforos Mavrogiannis, Jonathan A DeCastro, and Siddhartha S Srinivasa. Abstracting road traffic via topological braids: Applications to traffic flow analysis and distributed control. *The International Journal of Robotics Research*, 2023.
- [8] N. Funk, C. Schaff, R. Madan, T. Yoneda, J. U. De Jesus, J. Watson, E. K. Gordon, F. Widmaier, S. Bauer, S. S. Srinivasa, T. Bhattacharjee, M. R. Walter, and J. Peters. Benchmarking structured policies and policy optimization for real-world dexterous object manipulation. *IEEE Robotics and Automation Letters*, 7(1):478–485, 2022.
- [9] C. Kessens, M. Kaplan, T. Rocks, P.R. Osteen, J. Rogers, E. Stump, A. Hurwitz, J. Fink, L. Quang, M. Gonzalez, J. Patel, M. DiBlasi, S. Patel, M. Weiker, D. Patel, J. Bowkett, R. Detry, S. Karumanchi, L. Matthies, J. Burdick, Y. Oza, A. Agarwal, A. Dornbush, D. Saxena, M. Likhachev, K. Schmeckpeper, K. Daniilidis, A. Kamat, Sanjiban Choudhury, A. Mandalika, and S.S. Srinivasa. Human-scale mobile manipulation using RoMan. *Journal of Field Robotics*, 2:1232–1262, 2022.
- [10] J. Spencer, Sanjiban Choudhury, M. Barnes, M. Schmittle, M. Chiang, P. Ramadge, and S.S. Srinivasa. Expert intervention learning: An online framework for robot learning from explicit and implicit human feedback. *Autonomous Robots*, 46:99–113, 2022.
- [11] M. Bhardwaj, Sanjiban Choudhury, B. Boots, and S. Srinivasa. Leveraging Experience in Lazy Search. *Autonomous Robots*, 45(7):979–996, 2021.
- [12] J.G. Gammell, T. Barfoot, and S.S. Srinivasa. Batch Informed Trees (BIT*): Informed asymptotically optimal anytime search. *The International Journal of Robotics Research*, 2020.
- [13] B. Yang, P.E. Lancaster, S.S. Srinivasa, and J.R. Smith. Benchmarking robot manipulation with the rubik’s cube. *IEEE Robotics and Automation Letters*, 2020.
- [14] T. Bhattacharjee, G. Lee, H. Song, and S.S. Srinivasa. Towards robotic feeding: Role of haptics in fork-based food manipulation. *IEEE Robotics and Automation Letters*, 2019.
- [15] M. Chen, S. Nikolaidis, H. Soh, D. Hsu, and S.S. Srinivasa. Trust-aware decision making for human-robot collaboration: Model learning and planning. *ACM Transactions on Human-Robot Interaction*, 2019.
- [16] R. Holladay, O. Salzman, and S.S. Srinivasa. Minimizing task-space fréchet error via efficient incremental graph search. *IEEE Robotics and Automation Letters*, 2019.

- [17] Jeongseok Lee, Michael X. Grey, Sehoon Ha, Tobias Kunz, Sumit Jain, Yuting Ye, Siddhartha S. Srinivasa, Mike Stilman, and C. Karen Liu. DART: Dynamic animation and robotics toolkit. *The Journal of Open Source Software*, 3(22):500, feb 2018.
- [18] J.G. Gammell, T. Barfoot, and S.S. Srinivasa. Informed sampling for asymptotically optimal path planning. *IEEE Transactions on Robotics*, 34(4):966–984, 2018.
- [19] S. Javdani, H. Admoni, S. Pellegrinelli, S.S. Srinivasa, and J.A. Bagnell. Shared autonomy via hindsight optimization for teleoperation and teaming. *The International Journal of Robotics Research*, 37(7):717–742, 2018.
- [20] S. Nikolaidis, M. Kwon, J. Forlizzi, and S.S. Srinivasa. Planning with verbal communication for human-robot collaboration. *ACM Transactions on Human-Robot Interaction*, 7(3), 2018.
- [21] B. Calli, A. Singh, J. Bruce, W. W. Aaron, K. Konolige, S.S. Srinivasa, P. Abbeel, and A.M. Dollar. YCB benchmarking project: Object set, data set and their applications. *Journal of The Society of Instrument and Control Engineers*, 56(10):792–797, 2017.
- [22] I. Dryanovski, M.K. Klingensmith, S.S. Srinivasa, and J. Xiao. Large-scale, real-time 3D scene reconstruction on a mobile device. *Autonomous Robots*, 41(6):1423–1445, 2017.
- [23] M. Ghorbel, J. Pineau, R. Gourdeau, S. Javdani, and S.S. Srinivasa. A decision-theoretic approach for the collaborative control of a smart wheelchair. *International Journal of Social Robotics*, pages 1–15, 2017.
- [24] S. Nikolaidis, D. Hsu, and S.S. Srinivasa. Human-robot mutual adaptation in collaborative tasks: Models and experiments. *The International Journal of Robotics Research*, 36(5-7):618–634, 2017.
- [25] A. Johnson, J. King, and S.S. Srinivasa. Convergent planning. *IEEE Robotics and Automation Letters*, 2016.
- [26] M. Klingensmith, S.S. Srinivasa, and M. Kaess. Articulated robot manipulator simultaneous localization and mapping (ARM-SLAM). *IEEE Robotics and Automation Letters*, 2016.
- [27] M.C. Koval, N.S. Pollard, and S.S. Srinivasa. Pre- and post-contact policy decomposition for planar contact manipulation under uncertainty. *The International Journal of Robotics Research*, 35(1–3):244–264, 2016.
- [28] B. Calli, A. Walsman, A. Singh, S.S. Srinivasa, P. Abbeel, and A.M. Dollar. Benchmarking in manipulation research: Using the Yale-CMU-Berkeley object and model set. *IEEE Robotics and Automation Magazine*, 22(3):36–52, 2015.
- [29] A. Collet, B. Xiong, C. Gurau, M. Hebert, and S.S. Srinivasa. HerbDisc: Towards lifelong robotic object discovery. *The International Journal of Robotics Research*, 34(1):3–25, 2015.
- [30] A.D. Dragan, R. Holladay, and S.S. Srinivasa. Deceptive robot motion: synthesis, analysis and experiments. *Autonomous Robots*, 39(3):331–345, 2015.
- [31] M.C. Koval, N.S. Pollard, and S.S. Srinivasa. Pose estimation for planar contact manipulation with manifold particle filters. *The International Journal of Robotics Research*, 34(7):922–945, 2015.
- [32] A. Stentz, H. Herman, A. Kelly, E. Meyhofer, G.C. Haynes, D. Stager, B. Zajac, J.A. Bagnell, J. Brindza, C. Dellin, M. George, J. Gonzalez-Mora, S. Hyde, M. Jones, M. Laverne, M. Likhachev, L. Lister, M.D. Powers, O. Ramos, J. Ray, D.P. Rice, J. Scheffle, R. Sidki, S.S. Srinivasa, K. Strabala, J.P. Tardif, J. Valois, J.M. Vandeweghe, M.D. Wagner, and C. Wellington. CHIMP, the CMU highly intelligent mobile platform. *Journal of Field Robotics*, 32(2):209–228, 2015.
- [33] A.D. Dragan and S.S. Srinivasa. Integrating human observer inferences into robot motion planning. *Autonomous Robots*, 37(4):351–368, 2014.
- [34] R. Paolini, A. Rodriguez, S.S. Srinivasa, and M.T. Mason. A data-driven statistical framework for post-grasp manipulation. *The International Journal of Robotics Research*, 33(4):600–615, 2014.
- [35] M.R. Dogar, M.C. Koval, A. Tallavajhula, and S.S. Srinivasa. Object search by manipulation. *Autonomous Robots*, 36(1–2):153–167, 2013.
- [36] M.R. Dogar and S.S. Srinivasa. Physics-based manipulation in human environments. *Journal of the Robotics Society of Japan*, 31(4):353–357, 2013.

- [37] A.D. Dragan, K.T. Lee, and S.S. Srinivasa. Teleoperation with intelligent and customizable interfaces. *Journal of Human-Robot Interaction*, 1(3), 2013.
- [38] A.D. Dragan and S.S. Srinivasa. A policy-blending formalism for shared control. *The International Journal of Robotics Research*, 32(7):790–805, 2013. (Conference version was **Best Conference Paper Award Finalist, RSS 2012**).
- [39] K. Strabala, M.K. Lee, A.D. Dragan, J. Forlizzi, S.S. Srinivasa, M. Cakmak, and V. Micelli. Towards seamless human-robot handovers. *Journal of Human-Robot Interaction*, 2(1), 2013.
- [40] M. Zucker, R. Ratliff, A.D. Dragan, M. Pivtoraiko, M. Klingensmith, C. Dellin, J.A. Bagnell, and S.S. Srinivasa. CHOMP: Covariant Hamiltonian Optimization for Motion Planning. *The International Journal of Robotics Research*, 32(9–10):1164–1193, 2013.
- [41] M.R. Dogar and S.S. Srinivasa. A planning framework for non-prehensile manipulation under clutter and uncertainty. *Autonomous Robots*, 33(3):217–236, 2012. (Conference version was **Best Conference Paper Award Finalist, IEEE IROS 2010**).
- [42] R.A. Knepper, S.S. Srinivasa, and M.T. Mason. Toward a deeper understanding of motion alternatives via an equivalence relation on local paths. *The International Journal of Robotics Research*, 31(2):168–187, 2012.
- [43] M.T. Mason, A. Rodriguez, S.S. Srinivasa, and A.S. Vazquez. Autonomous manipulation with a general-purpose simple hand. *The International Journal of Robotics Research*, 31(5):688–703, 2012.
- [44] S.S. Srinivasa, D. Berenson, M. Cakmak, A. Collet, M.R. Dogar, A.D. Dragan, R.A. Knepper, T. Niemueller, K. Strabala, M. Vandeweghe, and J. Ziegler. HERB 2.0: Lessons learned from developing a mobile manipulator for the home. *Proceedings of the IEEE*, 100(8):1–19, 2012.
- [45] D. Berenson, S.S. Srinivasa, and J. Kuffner. Task Space Regions: A framework for pose-constrained manipulation planning. *The International Journal of Robotics Research*, 30(12):1435–1460, 2011.
- [46] A. Collet, M. Martinez, and S.S. Srinivasa. The MOPED framework: Object recognition and pose estimation for manipulation. *The International Journal of Robotics Research*, 30(10):1284–1306, 2011. (Conference version was **Best Vision Paper Award Finalist, IEEE ICRA 2009**).
- [47] S.S. Srinivasa, D. Ferguson, C.J. Helfrich, D. Berenson, A. Collet, R. Diankov, G. Gallagher, G. Hollinger, J. Kuffner, and M.V. Weghe. HERB: A Home Exploring Robotic Butler. *Autonomous Robots*, 28(1):5–20, 2010.
- [48] P. Yang, R.A. Freeman, G.J. Gordon, K.M. Lynch, S.S. Srinivasa, and R. Sukthankar. Decentralized estimation and control of graph connectivity for mobile sensor networks. *Automatica*, 46(2):390–396, 2010.
- [49] S.S. Siddhartha, R. Narasimha, A.J. Basu, and S.V. Kailas. Coherent structures in numerically simulated jets with and without off-source heating. *Fluid Dynamics Research*, 26(2):105–117, 2000.

Refereed Conferences

- [50] Jose Jaime, Liander Rainbolt, Ikshita Sathanur, Taylor Kessler Faulkner, and Siddhartha Srinivasa. Extending robot-assisted feeding to spoon-based foods with the articutool: A modular active end-effector. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026.
- [51] Quinn Pfeifer, Ethan Pronovost, Paarth Shah, Khimya Khetarpal, Abhishek Gupta, and Siddhartha Srinivasa. Difference-aware retrieval policies for imitation learning. In *International Conference on Learning Representations*, 2026.
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Reports and Theses

- [243] Ethan Pronovost, Khimya Khetarpal, and Siddhartha Srinivasa. Robust intervention learning from emergency stop interventions, 2026.
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Seminars

Keynote @ International Conference on Automated Planning and Scheduling	2022
Carnegie Mellon University	2022
MIT	2020
Army Research Laboratories	2019
Carnegie Mellon University	2019
Northwestern	2018
Toyota Technological Institute at Chicago	2018
Georgia Tech	2018
Amazon	2018
Microsoft Research	2017
Carnegie Mellon	2017
Princeton	2017
University of Washington	2017
Harvard	2016
MIT	2016
National University of Singapore	2014
University of Pennsylvania	2011
National Taiwan University	2010
Indian Institute of Technology Madras	2010
West Penn Hospital	2007

Others: Too numerous to count.

Teaching

CSE 478 Autonomous Robotics
Paul G. Allen School for Computer Science & Engineering

Winter 2017-

Brand new undergraduate-level robotics course on robotics in the real world. The course covers state estimation (particle filters, motion models, sensor models etc), planning/control (search based planners, lattice based planners, trajectory following techniques etc), and perception and learning (object detection, learning from demonstrations etc.). Student teams implement algorithms on the RACECAR platform developed by Prof. Srinivasa for the course.

CSE 599 Advanced Robotics

Fall 2017

Paul G. Allen School for Computer Science & Engineering

Brand new graduate-level robotics course on motion planning algorithms. The course covers the Piano Movers Problem, sampling-based planning, minimum dispersion graphs, efficient search, lazy and anytime planning, planning under uncertainty with application to mobile manipulators and humanoid robots, with a focus on algorithmic foundations and theorem proving.

16-843 Manipulation Algorithms

Fall 2012-16

The Robotics Institute, Carnegie Mellon University

Brand new graduate-level robotics course on the theory and algorithms that enable robots to physically manipulate their world. The course covers the geometry of manipulation configuration spaces, motion planning in these spaces, synthesizing robust and stable grasps for dexterous hands, reconfiguring clutter, task-level planning of multi-stage manipulation, physics-based actions, and addressing perception and model uncertainty, with application to mobile manipulators and humanoid robots.

16-662 Robot Autonomy

Spring 2012-16

The Robotics Institute, Carnegie Mellon University

Brand new graduate-level robotics course on manipulation, motion planning, perception, navigation, and machine learning algorithms for mobile manipulators. The course covers theory and algorithms, and has a strong hands-on component where students implement their assignments and class projects on a real mobile manipulation platform.

16-741 Mechanics of Manipulation

Spring 2009

The Robotics Institute, Carnegie Mellon University

Co-taught with Matt Mason

Graduate-level robotics core course on model-based robotic manipulation. To develop techniques for rigid body mechanics, kinematic constraint, Coulomb friction, gravity, and impact, and apply these techniques to manipulation problems including picking and placing, parts orienting, assembly, and mobile manipulation.

Professional Activities

Board Member	Robotics: Science and Systems Foundation	2016-
Board Member	International Journal of Robotics Research (IJRR)	2023-
Editor	International Journal of Robotics Research (IJRR)	2014-2022
Editor	IEEE/RSJ IROS	2014-2016
Editor	A Roadmap for U.S. Robotics: From Internet to Robotics	2013
Guest Editor	IJRR, RSS Special Issue	2013
Guest Editor	Autonomous Robots, RSS Special Issue	2013
Guest Editor	IEEE RAM, Special Issue on Mobile Manipulation	2012
Associate Editor	IEEE/RSJ IROS	2011-2012
Associate Editor	IEEE ICRA	2010-2013

Selected Organization

Organizer	UW CSE MSR Summer Institute on Social Robotics	2018
General Chair	Robotics: Science and Systems (RSS)	2018
Program Chair	Robotics: Science and Systems (RSS)	2017
Organizer	Dagstuhl Seminar on Multimodal Manipulation Under Uncertainty	2015
Presentations Chair	IEEE IROS	2014
Chair	IEEE ICRA Best Manipulation Paper Award Committee	2013
Publications Chair	RSS	2013
Founding Program Chair	Robotics Track AAI	2012-2013
Senior Program Committee	AAAI	2012-2013
Founding Chair	IEEE RAS Technical Committee on Mobile Manipulation	2010-2012
Short Presentations Chair	RSS	2012
Area Chair	RSS	2011-2012

Workshops Organized: Too numerous to count.

Long-term Program Committees: International Conference on Human-Robot Interaction (HRI), International Conference on Automated Planning and Scheduling (ICAPS), Robotics: Science and Systems (RSS), Workshop on the Algorithmic Foundations of Robotics (WAFR), AAI Special Track on Physically Grounded AI.

University Service

Member	UW CSE Undergraduate Admissions Committee	2022-
Member	UW CSE Graduate Admissions Committee	2017-2022
Member	RI Curriculum Program Committee	2015-2017
Member	RI Faculty Hiring Committee	2015-2017
Chair	SCS ACM Doctoral Dissertation Award Committee	2014
Chair	RI Admissions Committee	2014
Member	RI Director Search Committee	2014
Member	SCS Student Teaching Award Committee	2014
Member	SCS Graduate Fellowship Committee	2012-2014
Member	RI Admissions Committee	2012-2015
Member	QoLT Director Search Committee	2012

Grants (excludes unrestricted gifts)

National Science Foundation (#0646448) <i>Methodology for Applying Haptic Robotics to Agile Manufacturing</i> ""Small Business Phase II	William Townsend-2009 co-PI, PI: 99,785
""Defense Advanced Research Projects Agency (#DARPA-BAA-12-39)"" <i>CHIMP: the CMU Highly Intelligent Mobile Platform</i> Robotics Challenge Track A	Tony Stenz-2012 co-PI, PI: 000""
""Office of Naval Research (#ONR N00014-24-S-B001)"" <i>Learning Multi-Step Dexterous Bimanual Fine Manipulation for Shipboard Maintenance and Urban Exploration</i> Long Range BAA for Navy and Marine Corps Science and Technology	2024-2027 PI
Amazon <i>Leveraging the Common-Sense of Large Language Models for Robotic Manipulation</i> Amazon Science Hub Research Award	2024-2025 PI
Defense Advanced Research Projects Agency (#HR0011-23-C-0150) <i>Aggressive, Resilient, High-speed Navigation in Off-road Terrain</i> Robotic Autonomy in Complex Environments with Resiliency	2023-2025 co-PI, PI: Byron Boots
""National Science Foundation (#2132848)""	2022-2025

Towards Efficient, Safe, and Personalized Caregiving Robots National Robotics Initiative	PI
""Office of Naval Research (#ONR N00014-22-1-2593)"" <i>SquadBot v2: High Performance Humanoid Robot for Urban Exploration</i> Long Range BAA for Navy and Marine Corps Science and Technology	2022-2023 co-PI, PI: IHMC
""National Science Foundation (#2007011)"" <i>A Formalism for Robot-Assisted Feeding while Adjusting to User Preferences</i> Human-Centered Computing	2020-2023 PI
National Science Foundation (#1839371) <i>Safe Imitation Learning for Robotics</i> Division of Mathematical Sciences, the Division of Computing and Communication Foundations	2018-2021 co-PI, PI: Zaid Harchaoui
""Office of Naval Research (#ONR N00014-16-R-BA01)"" <i>Enabling dexterous physics-based manipulation via a learning framework for shared autonomy</i> Long Range BAA for Navy and Marine Corps Science and Technology	2017-2020 PI
National Science Foundation (#1748582) <i>Learning Deep Sensorimotor Policies for Shared Autonomy</i> National Robotics Initiative	2017-2019 co-PI, PI: Sergey Levine
Robotics Collaborative Technology Alliance <i>Robust Outdoor Mobile Manipulation</i> T3	2017-2018 PI
Amazon <i>Data Efficient Policy Search for Reinforcement Learning</i> Amazon Research Award	2017-2018 PI
Richard King Mellon Foundation <i>Intelligent assistive technology for individuals with physical disabilities</i> Research Award	2016-2017 PI
""National Science Foundation (#1544797)"" <i>Learning control sharing strategies for assistive cyber-physical systems</i> Cyber-Physical Systems	2015-2018 co-PI, PI: Brenna Argall
Defense Advanced Research Projects Agency <i>An Architecture for Shared Autonomy via Optimal Control</i> Simplifying Complexity in Scientific Discovery (SIMPLEX)	2015-2016 PI
National Science Foundation (#1409003) <i>The Foundations of a Manipulation Repertoire</i> Robust Intelligence, Division of Information & Intelligent Systems	2014-2018 co-PI, PI: Matt Mason
""National Institute of Health R01 (#R01EB019335)"" <i>A Formalism for Customizing and Training Intelligent Assistive Devices</i> Smart and Connected Health	2014-2017 co-PI, PI: Brenna Argall
Office of Naval Research (#ONR BAA 13-0001) <i>Mental Simulation of Intentions for Collaborative Human-Robot Learning and Planning</i> ONR Basic Research Challenges in the Science of Autonomy	2014-2017 co-PI, PI: Andrea Thomaz
Toyota <i>Physics-based Intelligent Manipulation in Clutter</i>	2013-2017 PI

Toyota Motor Engineering & Manufacturing

ABB
In-hand manipulation with a simple gripper
Research Grant
2013-2014
co-PI, PI: Matt Mason

Research for Advanced Manufacturing in Pennsylvania
Robotics-enhanced, Cost-effective Motion Test Equipment for Inertial MEMS Devices
Research Grant
2013-2014
co-PI, PI: David Bourne

Office of Naval Research
Enabling Advanced Autonomous Physical Manipulation Capabilities for Robots in Human-Robot Teams
Young Investigator Award (ONR-YIP)
2012-2015
PI

Intel
Lifelong Learning in the Real World
Embedded Computing Science and Technology Center
2012-2014
PI

""National Science Foundation (#1208388)""
Addressing Clutter and Uncertainty for Robotic Manipulation in Human Environments
National Robotics Initiative
2012-2013
PI

National Science Foundation (#1228906)
Building Intelligent Mobile Manipulators for Assistive Care
Robust Intelligence, Division of Information & Intelligent Systems
2012-2013
co-PI, PI: Matt Mason

Defense Advanced Research Projects Agency (#DARPA-BAA-10-28)
ANDY: Learning for Autonomous Robotic Manipulation
Autonomous Robotic Manipulation Software Track
2011-2013
co-PI, PI: Drew Bagnell

Defense Advanced Research Projects Agency (#DARPA-BAA-10-28)
REARM: Robust Extensible Autonomous Robotic Manipulation
Autonomous Robotic Manipulation Software Track
2010-2011
co-PI, PI: SRI

National Science Foundation (#0916557)
A Simple but General Hand
Robust Intelligence, Division of Information & Intelligent Systems
2009-2013
co-PI, PI: Matt Mason

National Science Foundation (#0540865)
Quality of Life Technology Engineering Research Center
Engineering Research Centers
2006-2016
co-PI, PI: Takeo Kanade

HONDA
Formalizing Mathematical Models of Curiosity
HONDA Research Institute
000""-
co-PI, PI: 700

US Army Research Laboratory
Safe, Fluent, and Generalizable Outdoor Autonomy
""Scalable
""150-000""
co-PI, PI: and Resilient Autonomy""

Selected Press Coverage ([Longer list](#))

[Video: The UW's assistive-feeding robot gets tested outside the lab](#) UW News, 2025
[Student Minigrant Story: Assistive-Feeding Robot Tested in the Real World](#) UW CREATE, 2025
[Robotics pioneer Sidd Srinivasa on the 'last mile problem,' humanoid hype, and why he's joining Madrona](#) Geek-

Wire, 2025

People with motor impairments help develop robotic feeding assistant at University of Washington Oregon Public Broadcast, 2025

Good Grips

Why Cruise AVs want everyone to win on the road

Amazon robotics leader leaves to join self-driving vehicle venture Cruise

Robotics Expert Sidd Srinivasa on Trends and What's Ripe for Innovation

New Science Hub to launch at Univ. of Washington with \$1.9M from Amazon

Warehouse Robotics Provider Berkshire Grey to Go Public Through \$2.7B SPAC Deal

These Small Cars Can Help Drive the Autonomous Future

The best interactive design of the year

Robot arm can feed people with mobility issues

UW's HERB robot makes cameo on X-Files as automated sushi waiter

Amazon hires top UW computer science prof as new robotics director

Siddhartha Srinivasa and Tao Xie named Fellows of the IEEE

Robotics expert moves entire team to UW, including famous Oreo-cracking robot

Learning to love our robot co-workers

Come on, Let's Give the Robots Hands Already

Uber Would Like to Buy Your Robotics Department

Robots that Care: The QoLT Center Changing the World of Caregiving

Robots 3D IMAX Movie

HERB: A robot that can unload a dishwasher and (sometimes) take apart an Oreo

Going Deep with David Rees: How to open a door

When it's too hard to separate Oreos ...

Helper Robots Are Steered, Tentatively, to Care for the Aging

HERB the Robot Butler Microwaves Your Dinner For You

What I think about when I think about robots

Can Robots Be Programmed to Learn from Their Own Experiences?

UW Engineering, 2025

LinkedIn, 2023

GeekWire, 2023

Madrona, 2022

GeekWire, 2022

Wall Street Journal, 2021

Wired, 2019

Fast Company, 2019

BBC News, 2019

GeekWire, 2018

GeekWire, 2018

Allen School News, 2017

GeekWire, 2017

New York Times, 2017

Wired, 2016

New York Times, 2015

NPR, 2014

National Geographic, 2014

Washington Post, 2014

National Geographic, 2014

CNN, 2013

New York Times, 2013

Popular Science, 2012

Granta, 2009

Scientific American, 2009

Extracurricular

- 140/13,072=0.01-th Place overall and 12/790=0.01-th Place in age group, Pittsburgh Half Marathon, 2017
- 6th Place overall and 1st Place in age group, Montour Trail Half Marathon, 2016
- 3rd Place, Finish MS 5M Run, 2015
- Fastest advisor-student time, Random Distance Run, 2013
- LaSalle Bank Chicago Marathon 2007, 2008
- Pittsburgh AB Squash League champion 2005
- Pittsburgh C Squash League champion 2002
- Institute silver medalist in Badminton, Indian Institute of Technology Madras 1998
- Institute bronze medalist in Tennis, Indian Institute of Technology Madras 1997, 1998